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# 04 - User Research Methods

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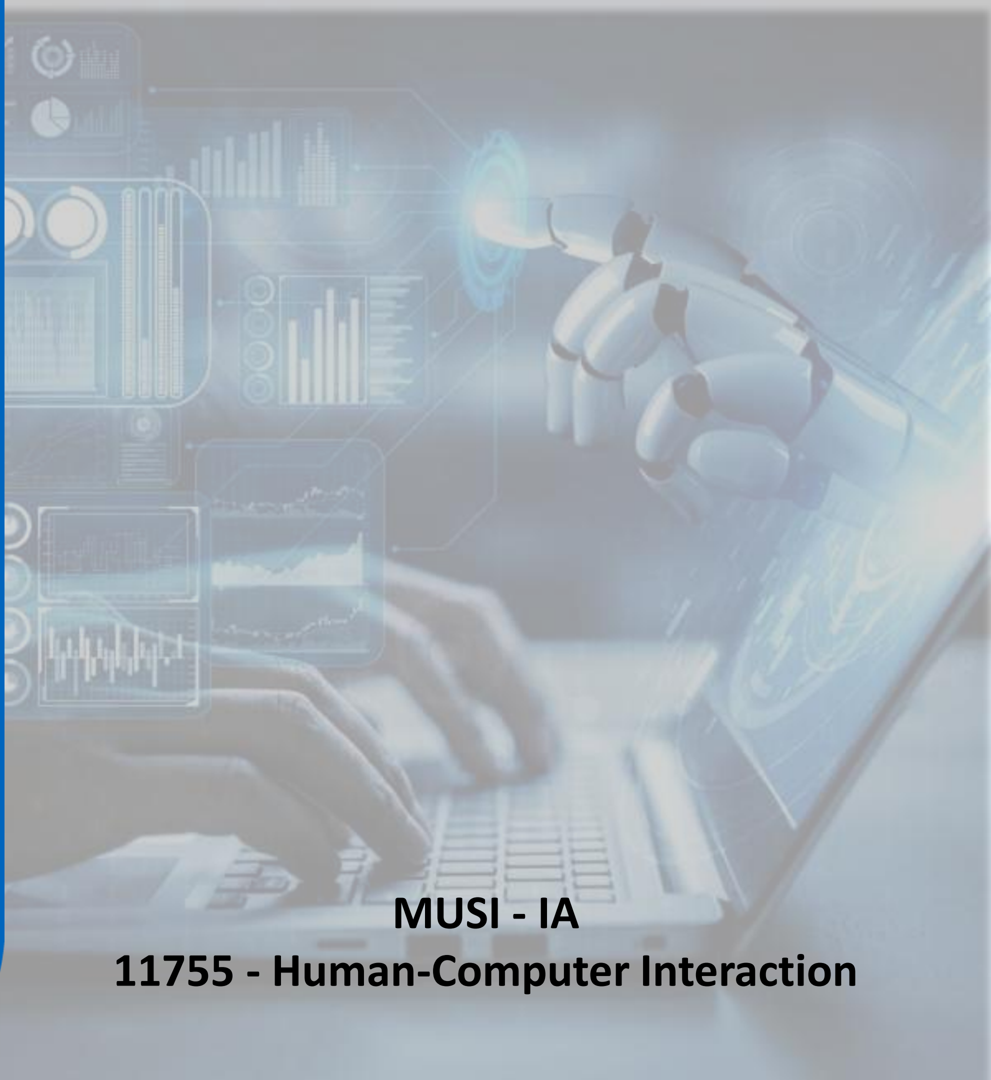

MUSI - IA

11755 - Human-Computer Interaction

# User Research in Design Thinking for Explainable AI Dashboards

- User research is absolutely crucial within the **Design Thinking process** when developing an **Explainable Artificial Intelligence (XAI)** dashboard.
- XAI, by its very nature, aims to make complex AI systems understandable to humans.
- However, "understandable" is subjective and depends heavily on the user.
- Without user research, we risk building XAI dashboards that are technically sound but fail to meet the actual needs and expectations of the intended users.

# The 5 Stages of the Design Thinking Process

The five stages of design thinking are:

- **Empathize:** research your users' needs.
- **Define:** state your users' needs and problems.
- **Ideate:** challenge assumptions and create ideas.
- **Prototype:** start to create solutions.
- **Test:** try your solutions out.

Interaction Design Foundation - IxDF. [“The 5 Stages in the Design Thinking Process”]

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# 1 User Research in Design Thinking for XAI

# Design Thinking Stage 1 - Empathize

**Purpose:** To gain a deep understanding of the users: their needs, goals, pain points, existing workflows, technical expertise, and expectations regarding AI explanations.

**User Research Methods:**

- **User Interviews:** Conducting structured and semi-structured interviews with various user groups (e.g., data scientists, business analysts, domain experts, end-users of AI-driven decisions). Focus on:
  - **What questions do they have about AI models?**
  - **What level of detail do they need in explanations?**
  - **What tasks do they perform related to AI outputs?**
  - **What are their current methods for understanding AI (if any)?**
  - **What are their frustrations with current AI systems?**
  - **What kind of explanations would build trust and confidence in AI?**

# Design Thinking Stage 1 - Empathize

**User Research Methods:**

- **Surveys and Questionnaires:** Distributing surveys to a larger user base to gather quantitative data and broader trends regarding user needs and preferences for XAI dashboards. Focus on demographics, technical proficiency, and preferred explanation formats.
- **Contextual Inquiry/Observation:** Observing users in their natural work environment as they interact with AI systems or related tools. This helps identify real-world workflows, pain points, and unmet needs related to understanding AI outputs.
- **User Personas:** Creating representative user profiles based on the research findings. Personas help to focus the design efforts on specific user needs throughout the design process.

# Design Thinking Stage 2 - Define

**Purpose:** To synthesize the insights gathered during the Empathize stage and clearly define the problem the XAI dashboard needs to solve from the user's perspective. This involves identifying user needs, pain points, and opportunities for improvement.

**User Research Methods (Building on Empathize):**

- **Affinity Mapping/Thematic Analysis:** Organizing and grouping the data collected in the Empathize stage (interview transcripts, survey responses, observation notes) to identify recurring themes, patterns, and user needs related to XAI.
- **"How Might We" Questions:** Framing the identified user needs and pain points as "How Might We" questions. This encourages solution-oriented thinking focused on user needs.

# Design Thinking Stage 3 - Ideate

**Purpose:** To brainstorm a wide range of potential solutions to the defined problem. This stage is about exploring various approaches to XAI dashboard design, explanation types, visualizations, and functionalities.

**User Research Methods (Incorporating User Perspective):**

- **Co-creation Workshops:** Involving users (or user representatives) in brainstorming sessions to generate ideas for the XAI dashboard. This ensures ideas are grounded in user needs and perspectives.
- **Card Sorting & Tree Testing:** Using card sorting to understand how users categorize and organize information related to AI explanations. Tree testing can be used to evaluate the findability of information within potential dashboard structures.

# Design Thinking Stage 4 - Prototype

**Purpose:** To create tangible representations of the best ideas from the Ideate stage and test them with users to get early feedback. This stage is about iterative refinement based on user interactions.

**User Research Methods (Central to Prototyping):**

- **Usability Testing (with Prototypes):** Conducting usability testing sessions with representative users to evaluate the prototypes. Focus on:
  - **Effectiveness:** Can users achieve their tasks using the prototype (e.g., understand an explanation, identify key factors, debug a model)?
  - **Efficiency:** How quickly and easily can users achieve their tasks?
  - **Satisfaction:** How satisfied are users with the prototype's usability and explainability?

# Design Thinking Stage 4 - Prototype

**User Research Methods (Central to Prototyping):**

- **A/B Testing (of Prototype Features):** Comparing different design options or explanation formats (e.g., different visualizations, explanation styles) with users to determine which performs better in terms of usability and understandability.

# Design Thinking Stage 5 - Test

**Purpose:** To rigorously evaluate the more developed prototypes or near-final dashboards with users to validate the design, identify remaining usability issues, and gather feedback for final refinements.

**User Research Methods (Formal Evaluation):**

- **Usability Testing (High-Fidelity Prototypes or Beta Dashboards):** Conducting more formal usability testing with higher-fidelity prototypes or beta versions of the dashboard. This might involve task-based scenarios, performance metrics (time on task, error rates), and user satisfaction surveys.
- **Heuristic Evaluation (XAI Principles):** Having usability experts (and potentially XAI experts) evaluate the dashboard based on established usability heuristics and principles for effective XAI.

# Design Thinking Stage 5 - Test

**User Research Methods (Formal Evaluation):**

- **User Feedback Surveys & Questionnaires (Post-Testing):** Gathering structured feedback from users after testing sessions to quantify usability and satisfaction and identify areas for improvement.
- **Analytics & Usage Data (if possible):** If a beta version is deployed, tracking user interaction data (e.g., which explanation types are used most, time spent on different sections, navigation patterns) to understand real-world usage patterns and identify areas for optimization.

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# 1.1 Example Plan for Stages 1 and 2

# Example Plan for Stages 1 y 2

**Goal:** To gain deep insights into the users of the XAI dashboard — their goals, pain points, mental models, and what they need from AI explanations.

**Key Questions to Answer:**

- Who are the primary users of the dashboard?
- What decisions do users make with the help of the AI?
- What kind of explanations do users expect, and at what level of detail?
- What factors influence users' trust in AI decisions?
- What frustrations do users face with current AI systems or black-box models?

# Example Plan for Stages 1 y 2 – Step 1: Define User Groups

**Identify all relevant people interacting with the system:**

- End-users: Decision-makers relying on AI insights (e.g., doctors, loan officers, analysts).
- AI developers & data scientists: Build and fine-tune models, want system interpretability.
- Domain experts: Understand the problem space and want AI to enhance their expertise.
- Managers & policymakers: Need explanations for accountability and compliance.

# Example Plan for Stages 1 y 2 – Step 2: Research Methods

1. **User Interviews (Primary Method)**

- Conduct 1-on-1 interviews (3–5 participants per group).
- Ask open-ended questions like:
  - Can you describe a recent situation where you had to trust an AI system's decision?
  - What makes an AI decision feel trustworthy or suspicious to you?
  - When an AI system makes a mistake, what kind of explanation would help you understand why?

2. **Surveys & Questionnaires**

- Broaden research with structured questions to reach more users.
- Collect data on users' familiarity with AI, preferences for explanation complexity, and pain points.

# Example Plan for Stages 1 y 2 – Step 2: Research Methods

3. **Contextual Inquiry / Field Studies**

- Observe users interacting with existing systems to understand real-world workflows.
- Take note of where explanations are missing, confusing, or overwhelming.

4. **Empathy Mapping Workshop**

- Gather stakeholders for a collaborative workshop.
- Build an empathy map: What users say, think, do, and feel when using AI systems.

# Example Plan for Stages 1 y 2 – Step 3: Synthesize Insights

1. Affinity Diagramming:
   - Cluster findings into themes — trust factors, explanation preferences, pain points, etc.
2. User Personas:
   - Create detailed personas representing different user types, highlighting their needs, motivations, and frustrations.
3. Customer Journey Mapping:
   - Map out users' interactions with the AI system to identify key touchpoints where explanations are critical.

# Example Plan for Stages 1 y 2 – Step 4: Define Research Outcomes

- Clear understanding of users' explanation needs (e.g., visual summaries vs. detailed justifications).
- Insights into decision-making workflows and how explanations fit into the process.
- Preliminary list of design requirements for the dashboard (e.g., interactive explanations, transparency indicators)..

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# 2 Conducting interviews and surveys

# User Interviews

**Conducting effective user interviews is key to gathering valuable insights for your design process — especially for something complex like an XAI dashboard.**

**Recommended Steps:**

🎯 1. Define Your Goals

🧠 2. Plan the Logistics

📝 3. Prepare Your Questions

🎧 4. Create a Comfortable Atmosphere

🔍 5. Capture and Synthesize Insights

📘 6. Iterate & Refine

# User Interviews - 🎯 Define Your Goals

Be clear on what you want to learn from the interviews. For example:

- **Understand user needs and pain points** with current AI tools.
- **Explore decision-making processes** and how users interpret AI explanations.
- **Identify trust factors** that make users more (or less) confident in AI decisions

# User Interviews - 🧠 2. Plan the Logistics

- **Participants:** Pick users that represent your personas (e.g., risk analysts, data scientists, compliance officers).
- **Format:** Decide on in-person, remote, or asynchronous interviews.
- **Tools:** Use tools like Zoom, Microsoft Teams, or even Notion for notes.
- **Time:** 20–30 minutes is usually a sweet spot — enough for depth, not too long to tire users.
- **Team Roles:** One person leads the interview, another takes notes (if possible).

# User Interviews - 📝 Prepare Your Questions

Create a mix of open-ended and probing questions:

- **Background & Context:**
  - "Can you walk me through your typical workflow?"
  - "What tools do you currently use for decision-making?"
- **Challenges & Pain Points:**
  - "What frustrates you most about AI-based systems?"
  - "Have you ever felt confused or unsure about an AI decision?“
- **Decision-Making Process:**
  - "How do you justify your decisions to clients or regulators?"
  - "What information do you need to feel confident about a prediction?"

# User Interviews - 📝 Prepare Your Questions

- **Expectations for Explainability:**
  - "What kind of explanations would be most helpful for you?"
  - "Would you prefer visual explanations, text summaries, or interactive charts?“

👉 **Tip:** Start broad, then dig deeper with follow-ups like: “Can you tell me more?” or “Why is that important to you?”

# User Interviews - 🎧 Create a Comfortable Atmosphere

- **Set Expectations:** Tell participants the purpose of the interview and how the insights will be used.
- **Build Trust:** Emphasize there are no "right" or "wrong" answers — you’re learning from them.
- **Active Listening:** Let them talk! Resist the urge to fill silences or guide their answers.
- **Neutral Reactions:** Stay unbiased. Even if they criticize your idea, that's valuable feedback.

# User Interviews - 🔍 Capture and Synthesize Insights

- **Record the Session (with consent):** This frees you to focus on the conversation.
- **Take Notes:** Jot down quotes, pain points, and anything surprising.
- **Thematic Analysis:** After interviews, group insights into common themes (e.g., “Trust issues,” “Feature importance confusion”).
- **Affinity Mapping:** Use tools to visually cluster user needs.

# User Interviews - 📘 Iterate & Refine

Treat interviews as an ongoing cycle:

- **Start with a few users**, refine your questions, and interview more people.
- Use findings to **update your personas, ideate features**, and **prototype explanations**.
- Repeat the process as your product evolves to stay aligned with users' needs.

# Sample Interview Script

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**11755 – Human-Computer Interaction**
**Summer Semester 2025**

**User Interview Sample Script for XAI Dashboard Development**

## Introduction (5 mins)

1. **Greet & Build Rapport:**
   - "Hi [Participant's Name], thank you for joining today! I'm [Your Name], and I'm working on designing a dashboard that makes AI decisions more understandable for users like you."
2. **Explain the Purpose:**
   - "We want to learn how you currently make decisions, your challenges with AI tools, and what would make explanations more useful and trustworthy for you. There are no right or wrong answers — we're just here to learn from your experience."
3. **Consent & Recording:**
   - "Would it be okay if we record this session to help us analyze the findings later?"
4. **Time & Structure:**
   - "This will take about 20–30 minutes. We'll start with some background questions, explore your current workflows, and then dive into your expectations for explainable AI tools."

## Section 1: Understanding the User's Context (5 mins)

1. **Role & Responsibilities:**
   - "Can you tell me a bit about your role and what a typical day looks like?"
   - "What types of decisions do you make, and how do they impact your work or clients?"
2. **Experience with AI Tools:**
   - "Do you use any AI-powered tools or dashboards in your workflow? If yes, which ones?"
   - "How do you feel about the decisions those tools make? Do you trust them?"
3. **Pain Points & Frustrations:**
   - "What's the most frustrating part of using AI in your work?"
   - "Have you ever struggled to understand why an AI system made a specific decision?"

## Section 2: Decision-Making Process (5 mins)

1. **Gathering Information:**
   - "What kind of information do you usually rely on to make decisions?"
   - "If an AI model gave you a prediction, what would you need to see to feel confident in that prediction?"
2. **Explaining Decisions to Others:**
   - "Do you ever need to explain AI-driven decisions to clients, managers, or regulators?"
   - "If yes, what makes an explanation easier or harder to communicate?"

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**11755 – Human-Computer Interaction**
**Summer Semester 2025**

**Ejemplo de Guion de Entrevista para el Desarrollo del Panel de Control XAI**

## Introducción (5 mins)

1. **Saludo y creación de empatía:**
   - "Hola [Nombre del participante], ¡gracias por unirse hoy! Soy [Tu nombre], y estoy trabajando en diseñar un panel de control que haga que las decisiones de IA sean más comprensibles para usuarios como usted."
2. **Explicar el propósito:**
   - "Queremos aprender cómo toma decisiones actualmente, cuáles son sus desafíos con las herramientas de IA y qué haría que las explicaciones fueran más útiles y confiables para usted. No hay respuestas correctas o incorrectas, estamos aquí para aprender de su experiencia."
3. **Consentimiento y grabación:**
   - "¿Estaría bien si grabamos esta sesión para ayudarnos a analizar los hallazgos más adelante?"
4. **Tiempo y estructura:**
   - "Esto tomará aproximadamente entre 20 y 30 minutos. Comenzaremos con algunas preguntas de contexto, exploraremos su flujo de trabajo actual y luego profundizaremos en sus expectativas sobre las herramientas de IA explicable."

## Sección 1: Comprender el contexto del usuario (5 mins)

1. **Rol y responsabilidades:**
   - "¿Puede contarme un poco sobre su rol y cómo es un día típico para usted?"
   - "¿Qué tipo de decisiones toma y cómo impactan en su trabajo o en sus clientes?"
2. **Experiencia con herramientas de IA:**
   - "¿Utiliza alguna herramienta o panel de control impulsado por IA en su flujo de trabajo? Si es así, ¿cuáles?"
   - "¿Cómo se siente respecto a las decisiones que toman esas herramientas? ¿Confía en ellas?"
3. **Puntos de dolor y frustraciones:**
   - "¿Cuál es la parte más frustrante de usar IA en su trabajo?"
   - "¿Alguna vez le ha costado entender por qué un sistema de IA tomó una decisión específica?"

## Sección 2: Proceso de toma de decisiones (5 mins)

1. **Recolección de información:**
   - "¿En qué tipo de información suele confiar para tomar decisiones?"
   - "Si un modelo de IA le diera una predicción, ¿qué necesitaría ver para sentirse seguro con esa predicción?"
2. **Explicaciones a terceros:**
   - "¿Alguna vez necesita explicar las decisiones impulsadas por IA a clientes, gerentes o reguladores?"

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# 2.1 Thematic Analysis

# Thematic analysis

- **Thematic analysis is a qualitative research method used to identify, analyze, and interpret patterns (or themes) within a set of qualitative data — like interview transcripts, open-ended survey responses, or focus group discussions**.
- It's a flexible approach that helps researchers make sense of large amounts of text by organizing it into meaningful categories that reflect key ideas or insights.

# Thematic analysis – Why?

**It helps you:**

- Understand people's experiences, thoughts, and perspectives.
- Spot common patterns or recurring ideas across multiple participants.
- Summarize the most important aspects of your research data.
- Build a structured narrative that supports your research goals (e.g., designing an XAI dashboard).

# Thematic analysis - 6 Phases

1. **Familiarization with the Data:**
   - Transcribe and read through the data multiple times.
   - Take initial notes and get a general sense of the content.
2. **Generating Initial Codes:**
   - Break the data into smaller segments by labeling interesting or relevant elements (these are "codes").
   - For example, if a participant says, “I don’t trust the AI when I can’t see how it made a decision,” you might code this as trust, decision visibility, or explanation necessity.

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. Qualitative Research in Psychology, 3(2), 77–101. https://doi.org/10.1191/1478088706qp063oa

# Thematic analysis - 6 Phases

3. **Searching for Themes:**
   - Group related codes together to form broader themes.
   - For instance, codes like trust, explanation necessity, and model transparency might cluster under the theme of **"Trust in AI Decisions"**.
4. **Reviewing Themes:**
   - Refine the themes by checking if they accurately reflect the data.
   - Merge or split themes as needed to ensure they represent meaningful patterns.

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. Qualitative Research in Psychology, 3(2), 77–101. https://doi.org/10.1191/1478088706qp063oa

# Thematic analysis - 6 Phases

5. **Defining and Naming Themes:**
   - Finalize your themes and describe what each one captures.
   - Make sure each theme has a clear definition and tells a coherent story about the data.
6. **Writing the Report:**
   - Present the themes with supporting quotes and interpretations.
   - Explain how these themes answer your research question or inform your design process.

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. Qualitative Research in Psychology, 3(2), 77–101. https://doi.org/10.1191/1478088706qp063oa

# Thematic analysis – Example (for an XAI Dashboard)

- **Data:** User interview transcripts.
- **Codes:** frustration with black-box models, desire for control, interpretability, feature importance visualization.
- **Themes:**
  - **Trust and Transparency:** Users want clear explanations to build confidence in AI decisions.
  - **Decision Support, Not Automation:** Users prefer AI as a decision aid, not a replacement.
  - **Visualization of Model Logic:** Users find feature importance charts and decision trees useful for understanding outputs.

# Thematic analysis – Python Tools

**Transcription Tools:**

- **OpenAI Whisper** — State-of-the-art speech-to-text model for accurate transcription.
- **AssemblyAI** (API-based) — Powerful automatic speech recognition with features like speaker diarization.
- **SpeechRecognition** — Simple library for basic speech-to-text tasks, works with Google Speech API.

# Thematic analysis – Python Tools

**Thematic Analysis:**

- **NLTK (Natural Language Toolkit)** — Useful for tokenizing, lemmatizing, and extracting themes.
  - **Example:** Use FreqDist to find common words and phrases.
- **spaCy** — Fast and powerful NLP library for text processing and named entity recognition (NER).
  - **Example:** Extract key concepts, entities, and relationships.
- **BERTopic** — Topic modeling with transformer-based embeddings (great for clustering themes).
  - **Example:** Automatically group interview text into thematic topics.

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# 2.2 Affinity Mapping

# Affinity Mapping

- **Affinity mapping (or affinity diagramming) is a collaborative technique used to organize and categorize ideas, insights, or pieces of information into meaningful clusters based on their relationships or shared themes.**
- It’s especially useful in user research to make sense of complex qualitative data — like interview transcripts or user feedback — by visually grouping similar concepts together.
- It helps you see patterns, discover connections, and prioritize the most important insights!

# Affinity Mapping – Why?

- **Identify Patterns:** Spot recurring themes in user interviews or observations.
- **Synthesize Data:** Turn scattered pieces of feedback into clear, organized groups.
- **Prioritize Insights:** Highlight the most critical pain points or desires.
- **Foster Collaboration:** It's a great activity for teams to collectively make sense of messy data.

# Affinity Mapping – How?

1. **Collect Your Data:**
    - Gather all your research findings — interview quotes, survey answers, observation notes, etc.
    - Write each insight on a sticky note (or a digital card if using online tools).
2. **Cluster Similar Insights:**
    - Start grouping sticky notes with similar ideas.
    - No predefined categories — let clusters emerge naturally as you group notes.
    - For example, quotes like *“I don’t understand why the model chose this option”* and *“I need to know which factors influenced the prediction”* might cluster under **“Need for Explainability”**.

# Affinity Mapping – How?

3. **Label the Groups:**
   - Give each cluster a meaningful title that captures the core theme.
   - For instance, a group of notes about users feeling unsure of model accuracy could be labeled **“Trust & Confidence”**.
4. **Refine the Groupings:**
   - Merge, split, or reorganize clusters as needed.
   - If something doesn’t fit anywhere, create a separate "Miscellaneous" or "Outliers" group.
5. **Extract Insights & Prioritize:**
   - Analyze the clusters to identify high-level insights.
   - Prioritize the most frequently mentioned or impactful themes — these become your guiding design principles!

# Affinity Mapping – Example for an XAI Dashboard

| Cluster | Example Insights | Possible Dashboard Feature |
| --- | --- | --- |
| **Trust & Confidence** | “I want to know why the model made this choice.” “I don’t trust decisions without an explanation.” | Add feature importance charts and explanation logs |
| **Decision-Making Support** | “I’d like to test different scenarios.” “Can I see what happens if I change a parameter?” | Include interactive “what-if” analysis |
| **Ease of Understanding** | “The explanations are too technical.” “I need simpler, more visual insights.” | Provide visual summaries or decision flow diagrams |
| **Human-AI Collaboration** | “I want AI to assist, not decide for me.” “I prefer AI suggestions rather than automated actions.” | Allow users to override AI decisions or get hints |

# Affinity Mapping – How?

3. **Label the Groups:**
    - Give each cluster a meaningful title that captures the core theme.
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    - Analyze the clusters to identify high-level insights.
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# Affinity Mapping – Python Tools

- **NetworkX** — Create graphs to visualize relationships between themes.
  - **Example:** Build a network of connected concepts.
- **Matplotlib** / **Seaborn** — Plot heatmaps, bar charts, or cluster diagrams.
  - **Example:** Visualize the frequency of themes or concepts.
- **Plotly** — Interactive charts for more dynamic affinity maps.
  - **Example:** Drag-and-drop style visualizations of thematic clusters.
- **hdbscan** + **UMAP** — Cluster interview responses and reduce dimensionality for visualizing theme groupings

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# 3 User Personas

# User Personas

- **A user persona is a fictional, yet realistic, representation of a target user based on research and data**.
- It captures key characteristics, goals, needs, and pain points of a specific type of user, helping designers, developers, and researchers better understand and empathize with the people they’re building for.
- In essence, it answers the question: “Who are we designing this for?”

# User Personas – Key Elements

- **Name & Role:** A fictional name and job title to humanize the persona.
- **Demographics:** Age, location, education, experience, and background.
- **Goals & Motivations:** What the user wants to achieve with the product.
- **Pain Points & Challenges:** What obstacles or frustrations they face.
- **Behavior & Preferences:** How they interact with technology and make decisions.
- **Needs & Expectations:** What features or explanations they require to feel confident using the system.

# User Personas – Why?

- **Empathy:** Helps teams step into users' shoes and design with their needs in mind.
- **Focus:** Keeps design decisions user-centered, not just feature-driven.
- **Alignment:** Ensures everyone on the team shares a common understanding of the target audience.

For an Explainable AI (XAI) dashboard, you might have personas like:

- **The Risk Analyst** who needs clear explanations to justify loan decisions.
- **The Data Scientist** who wants in-depth feature importance for model tuning.
- **The Compliance Officer** who needs to trace decisions for regulatory audits.

# User Personas – Steps to Create One

1. **Conduct User Research**

- Gather data to understand who your users are, their behaviors, and their needs. You can use:
  - **User Interviews** — Talk to potential users directly.
  - **Surveys & Questionnaires** — Collect quantitative and qualitative insights.
  - **Observations & Usability Testing** — Watch how users interact with similar tools.
  - **Customer Feedback & Support Logs** — Identify recurring patterns and pain points.

# User Personas – Steps to Create One

2. **Identify Patterns & Group Insights**

- Once you've collected the data, start looking for commonalities:
  - **Goals:** What do users want to accomplish?
  - **Challenges:** What frustrates or blocks them?
  - **Behaviors:** How do they typically work, think, and make decisions?
  - **Technology Use:** Are they tech-savvy or beginners?

# User Personas – Steps to Create One

3. **Define the Persona Details**

- **Name & Picture:** Humanize the persona (e.g., Sarah the Risk Analyst).
- **Job Role & Industry:** Describe their profession and domain knowledge.
- **Demographics:** Age, location, education, experience, etc.
- **Goals & Motivations:** What are they trying to achieve with the product?
- **Pain Points & Frustrations:** What challenges do they face?
- **Behavior & Workflow:** How do they currently approach tasks?
- **Technology Comfort Level:** Are they comfortable with complex systems or need simplicity?

# User Personas – Steps to Create One

4. **Visualize & Share the Persona**

- Turn your insights into a one-page document or poster. Add quotes, scenarios, or a mini-bio to bring the persona to life. Share it with your team so everyone has a shared understanding of the users!

5. **Keep Iterating & Updating**

- Personas aren’t static — as you collect more data or your product evolves, revisit and refine them to stay aligned with real users.

# User Personas – Examples

**Persona 1:** Sarah, the Risk Analyst

- **Age:** 32
- **Job Role:** Risk Analyst at a financial institution
- **Tech Proficiency:** Moderate
- **Needs:** Understand why a loan application was denied or approved
- **Pain Points:** Complex models make it hard to justify decisions to clients
- **Goals:** Use AI explanations to streamline client reporting and reduce decision appeals

# User Personas – Examples

**Persona 2:** Mark, the Data Scientist

- **Age:** 28
- **Job Role:** Data Scientist at a healthcare startup
- **Tech Proficiency:** High
- **Needs:** Tune models based on explanations and feature importance
- **Pain Points:** Explanations aren’t always human-friendly
- **Goals:** Balance model accuracy with interpretability for non-technical stakeholders

# User Personas – Examples

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**11755 – Human-Computer Interaction**
**Summer Semester 2025**

### Sample User Persona for XAI Dashboard User Research

👤 **Name:** Sarah Mitchell

**Role: Risk Analyst** at a financial institution

**Industry:** Banking & Loan Approvals

**Primary Goal:** Understand why AI approves or rejects loan applications to make informed decisions and explain them to clients.

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#### Background & Context

1. 35 years old, 10+ years in finance.
2. Regularly reviews loan applications and uses AI-based recommendations.
3. Needs to **trust and explain AI decisions** to customers and regulators.
4. Has some data literacy but is **not a data scientist**—prefers **clear, actionable insights** over technical jargon.

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#### Pain Points & Challenges

- **Lack of transparency:** AI decisions feel like a "black box."
- **Complex explanations:** Wants **simple, visual summaries** rather than technical details.
- **Regulatory compliance:** Needs to justify approvals/rejections based on **explainability standards.**
- **Customer communication:** Struggles to explain **why a client was denied a loan** based on AI output.

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#### Needs & Expectations from the XAI Dashboard

- **Clear explanations:** Wants to see **why a decision was made** in **plain language + visuals.**
- **Feature importance:** Needs a **ranking of factors** influencing a decision (e.g., credit score vs. income).
- **Alternative scenarios:** Would benefit from **"What if?" analyses** (e.g., "What if the applicant had a higher income?").
- **Interactive UI:** Prefers a **dashboard with sliders** to test different conditions and see AI responses dynamically.

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#### Preferred Explanation Methods

- **Feature importance plots** (e.g., SHAP, LIME)
- **Counterfactual explanations** ("If X were different, the decision would change")
- **Decision flowcharts** (step-by-step reasoning behind a prediction)
- **Confidence scores with clear thresholds**

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